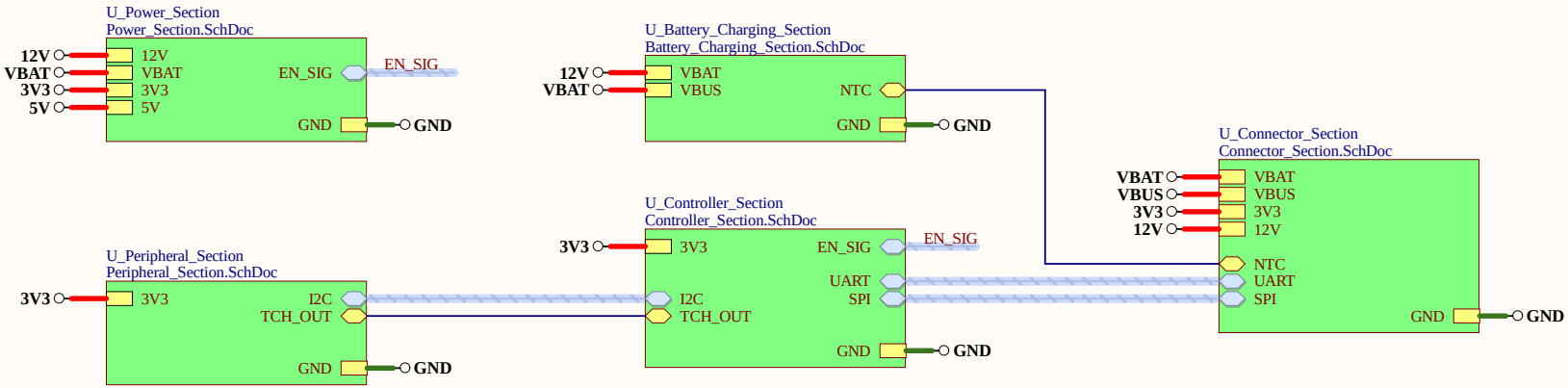
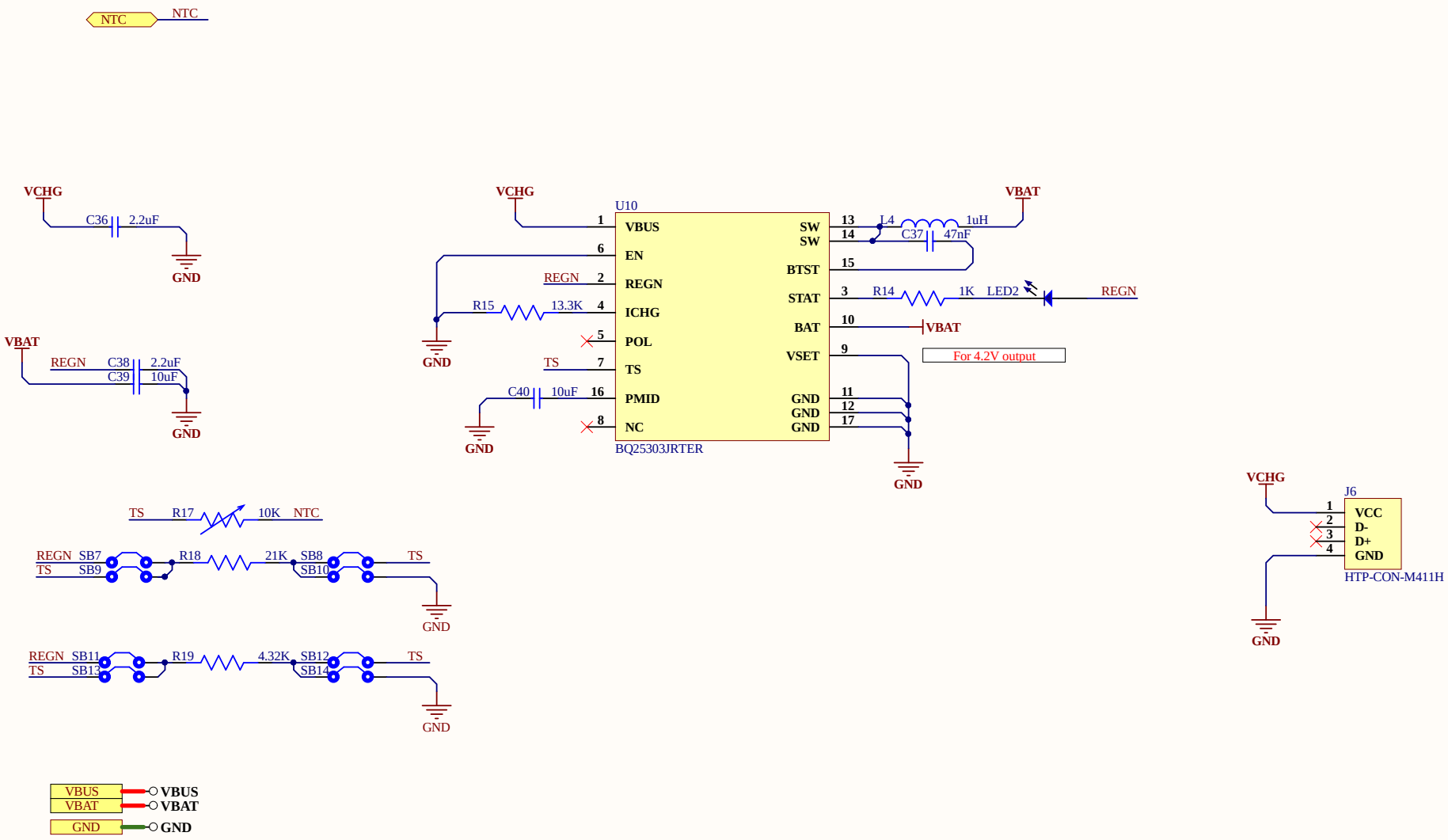
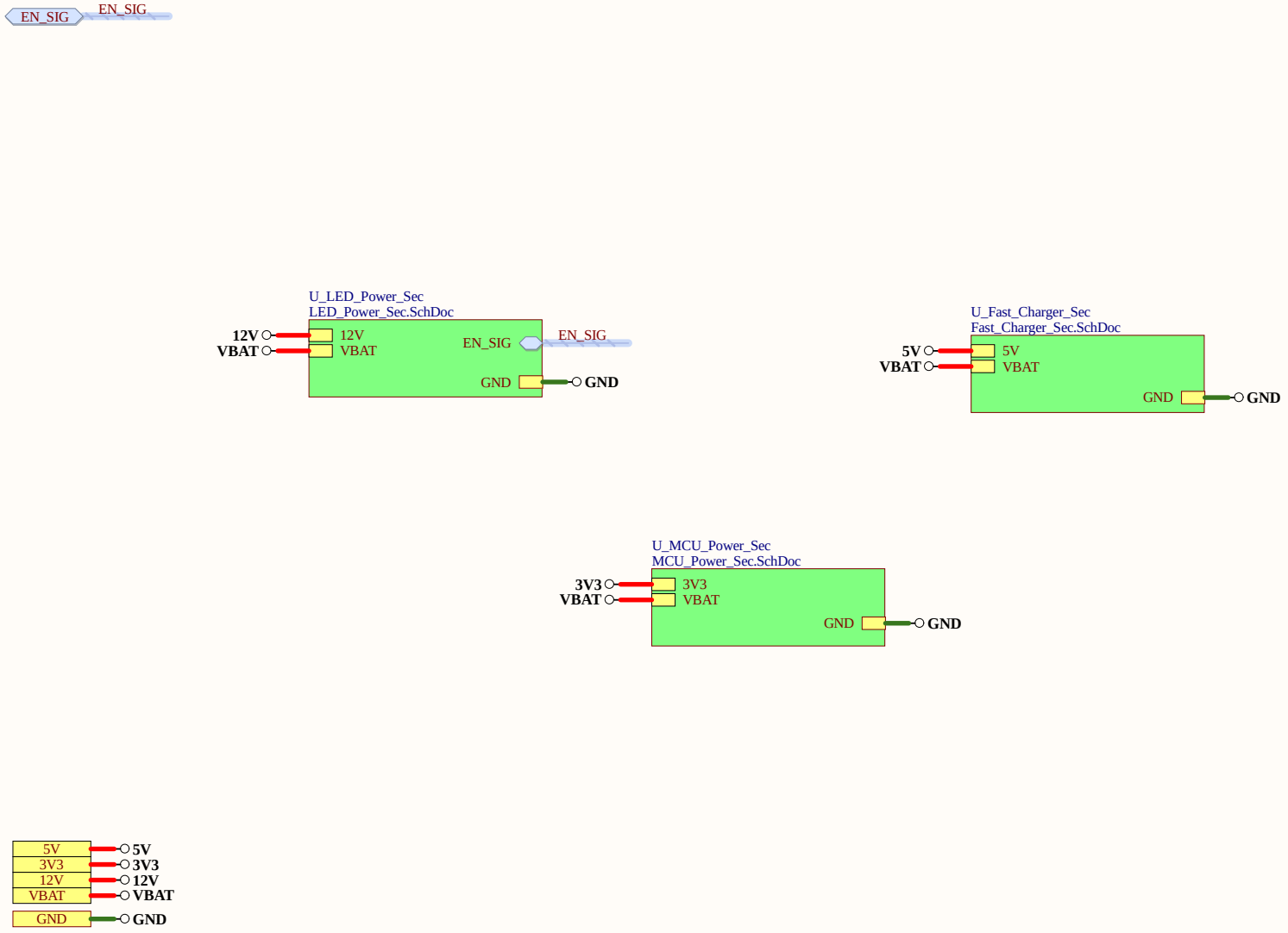
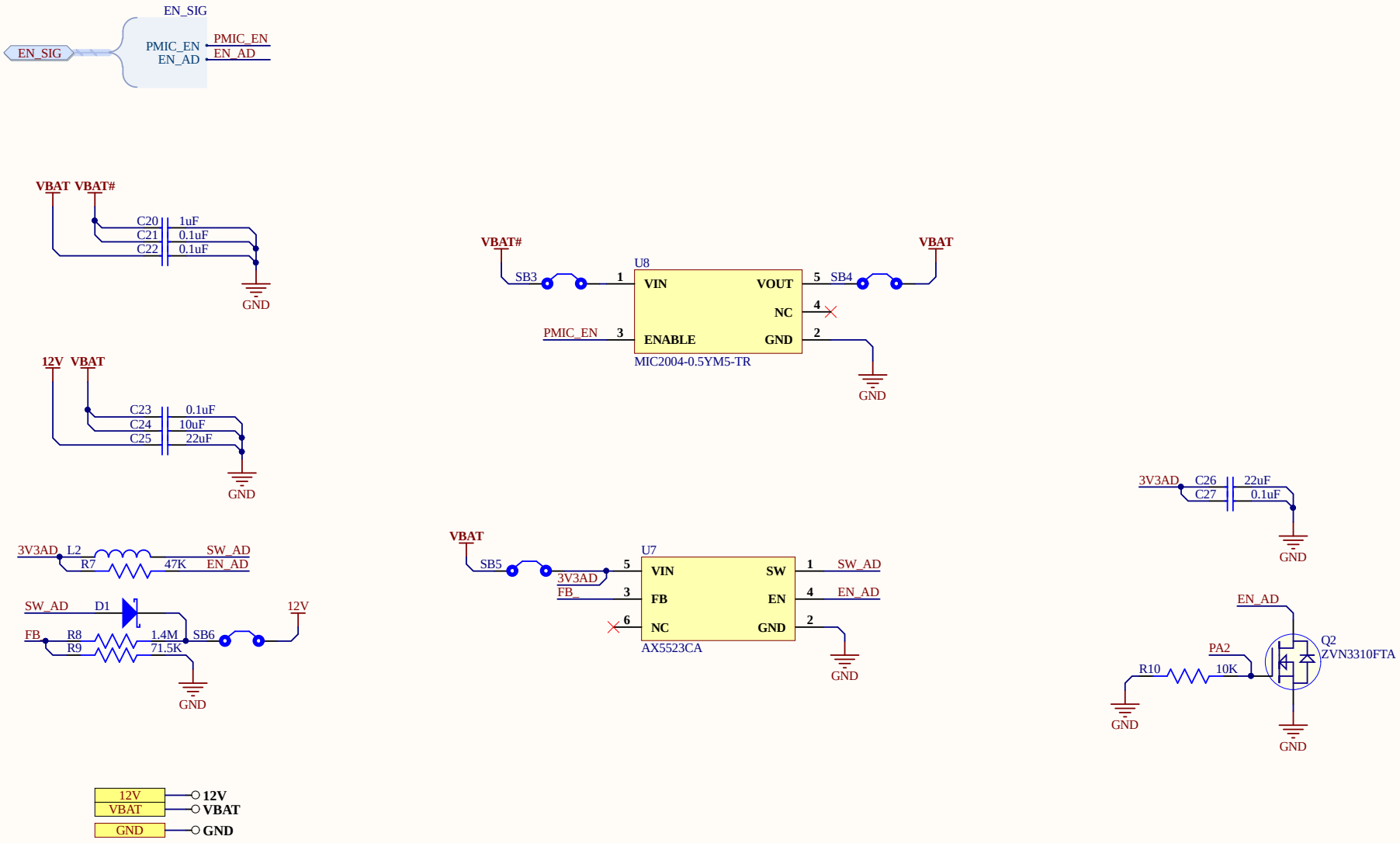
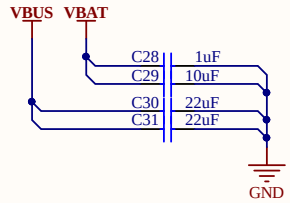
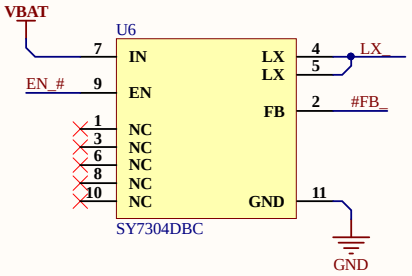

U_Power_Section
Power_Section.SchDoc
12V
VBAT
3V3
5V
12V
VBAT
3V3
5V
EN_SIG
EN_SIG
GND
GND
U_Battery_Charging_Section
Battery_Charging_Section.SchDoc
12V
VBAT
VBAT
VBUS
NTC
GND
GND
U_Connector_Section
Connector_Section.SchDoc
VBAT
VBUS
3V3
12V
VBAT
VBUS
3V3
12V
NTC
UART
SPI
GND
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
EN_SIG
EN_SIG
UART
SPI
I2C
TCH_OUT
I2C
TCH_OUT
GND
GND
U_Peripheral_Section
Peripheral_Section.SchDoc
3V3
3V3
I2C
TCH_OUT
GND
GND

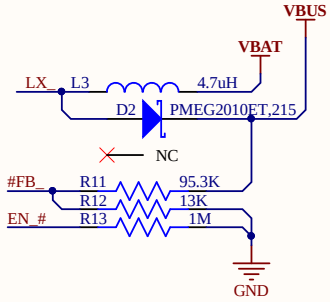

NTC
NTC
VCHG
C36
2.2uF
GND
VBAT
REGN
C38
2.2uF
C39
10uF
GND
VCHG
U10
1
VBUS
6
EN
REGN
2
REGN
R15
13.3K
4
ICHG
5
POL
TS
7
TS
C40
10uF
16
PMID
8
NC
GND
GND
BQ25303JRTER
SW
13
SW
14
BTST
15
STAT
3
BAT
10
VSET
9
GND
11
GND
12
GND
17
L4
1uH
C37
47nF
VBAT
R14
1K
LED2
REGN
VBAT
For 4.2V output
GND
TS
R17
10K
NTC
REGN
SB7
TS
SB9
R18
21K
SB8
SB10
TS
GND
REGN
SB11
TS
SB13
R19
4.32K
SB12
SB14
TS
GND
VCHG
J6
1
2
3
4
VCC
D-
D+
GND
HTP-CON-M411H
GND
VBUS
VBAT
GND
VBUS
VBAT
GND

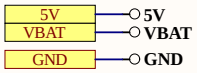

EN_SIG
EN_SIG
U_LED_Power_Sec
LED_Power_Sec.SchDoc
12V
VBAT
EN_SIG
GND
12V
VBAT
EN_SIG
GND
U_Fast_Charger_Sec
Fast_Charger_Sec.SchDoc
5V
VBAT
5V
VBAT
GND
GND
U_MCU_Power_Sec
MCU_Power_Sec.SchDoc
3V3
VBAT
3V3
VBAT
GND
GND
5V
3V3
12V
VBAT
GND
5V
3V3
12V
VBAT
GND

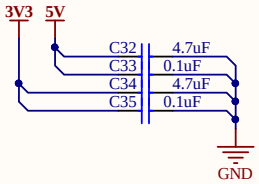

EN_SIG
EN_SIG
PMIC_EN
EN_AD
PMIC_EN
EN_AD
VBAT VBAT#
C20 1uF
C21 0.1uF
C22 0.1uF
GND
12V VBAT
C23 0.1uF
C24 10uF
C25 22uF
GND
3V3AD L2 SW_AD
R7 47K EN_AD
SW_AD D1 12V
FB R8 1.4M SB6
R9 71.5K
GND
12V
VBAT
GND
12V
VBAT
GND
VBAT# VBAT
U8
SB3 1 VIN VOUT 5 SB4
NC 4
PMIC_EN 3 ENABLE GND 2
MIC2004-0.5YM5-TR
GND
VBAT
U7
SB5 5 VIN SW 1 SW_AD
3V3AD
FB_ 3 FB EN 4 EN_AD
6 NC GND 2
AX5523CA
GND
3V3AD C26 22uF
C27 0.1uF
GND
EN_AD
PA2 Q2
R10 10K ZVN3310FTA
GND
GND

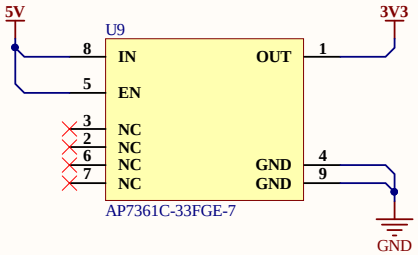

VBUS
VBAT
C28
1uF
C29
10uF
C30
22uF
C31
22uF
GND

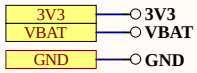

VBAT
U6
7
IN
EN_#
9
EN
1
NC
3
NC
6
NC
8
NC
10
NC
LX
4
LX
5
FB
2
#FB_
GND
11
SY7304DBC
GND

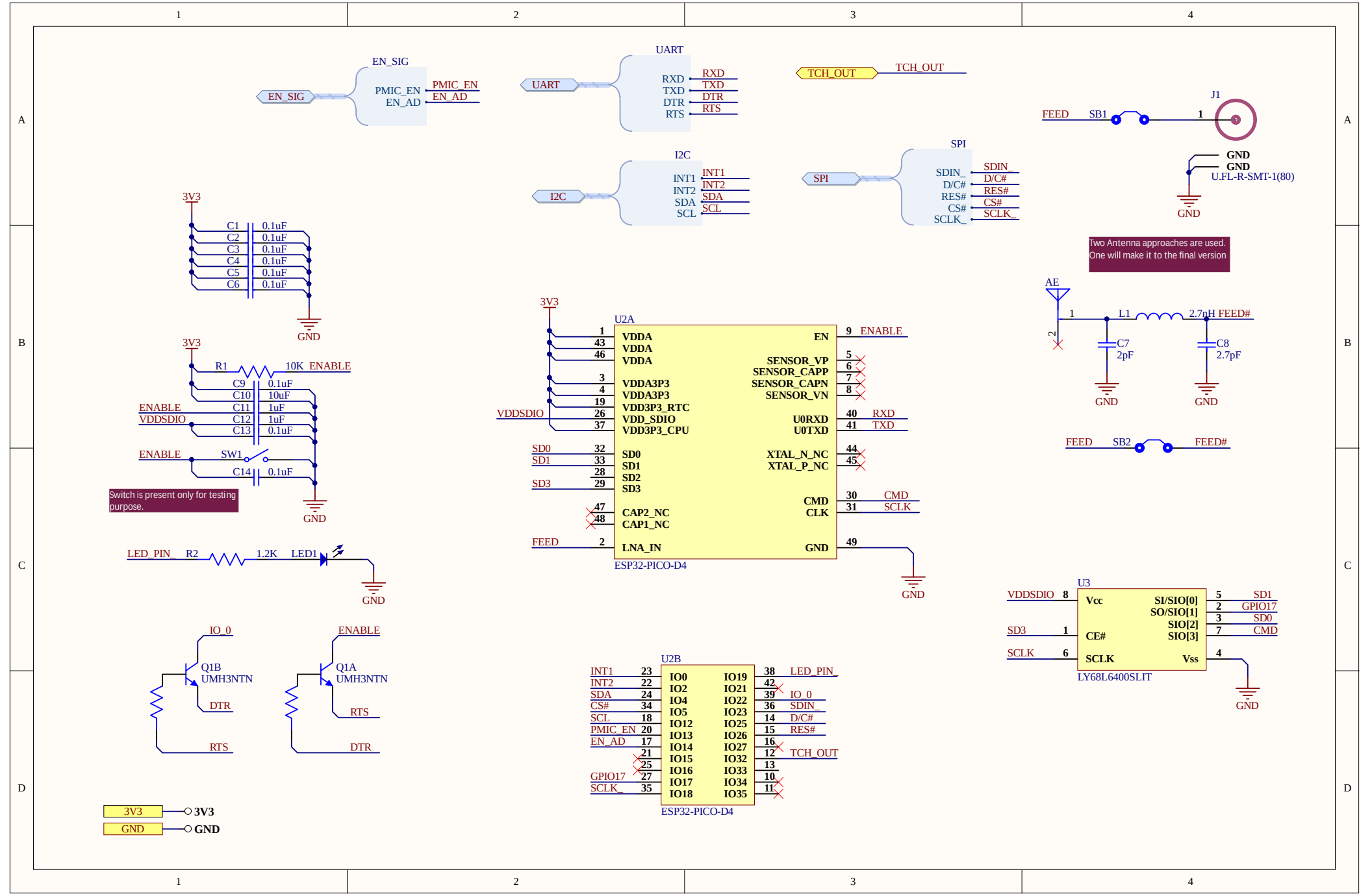

VBUS
VBAT
LX_
L3
4.7uH
D2
PMEG2010ET,215
NC
#FB_
R11
95.3K
R12
13K
EN_#
R13
1M
GND

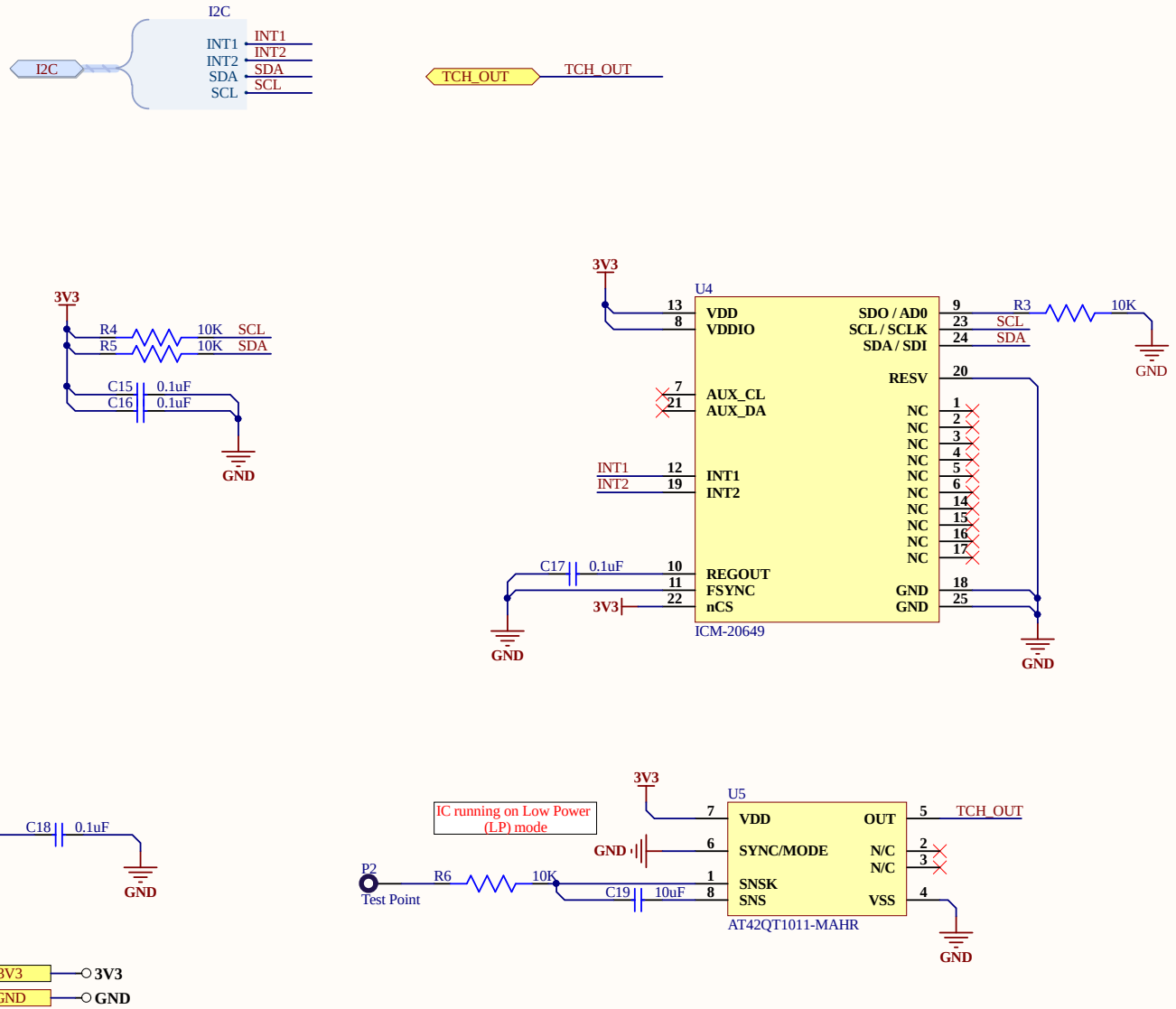

5V
VBAT
GND
5V
VBAT
GND

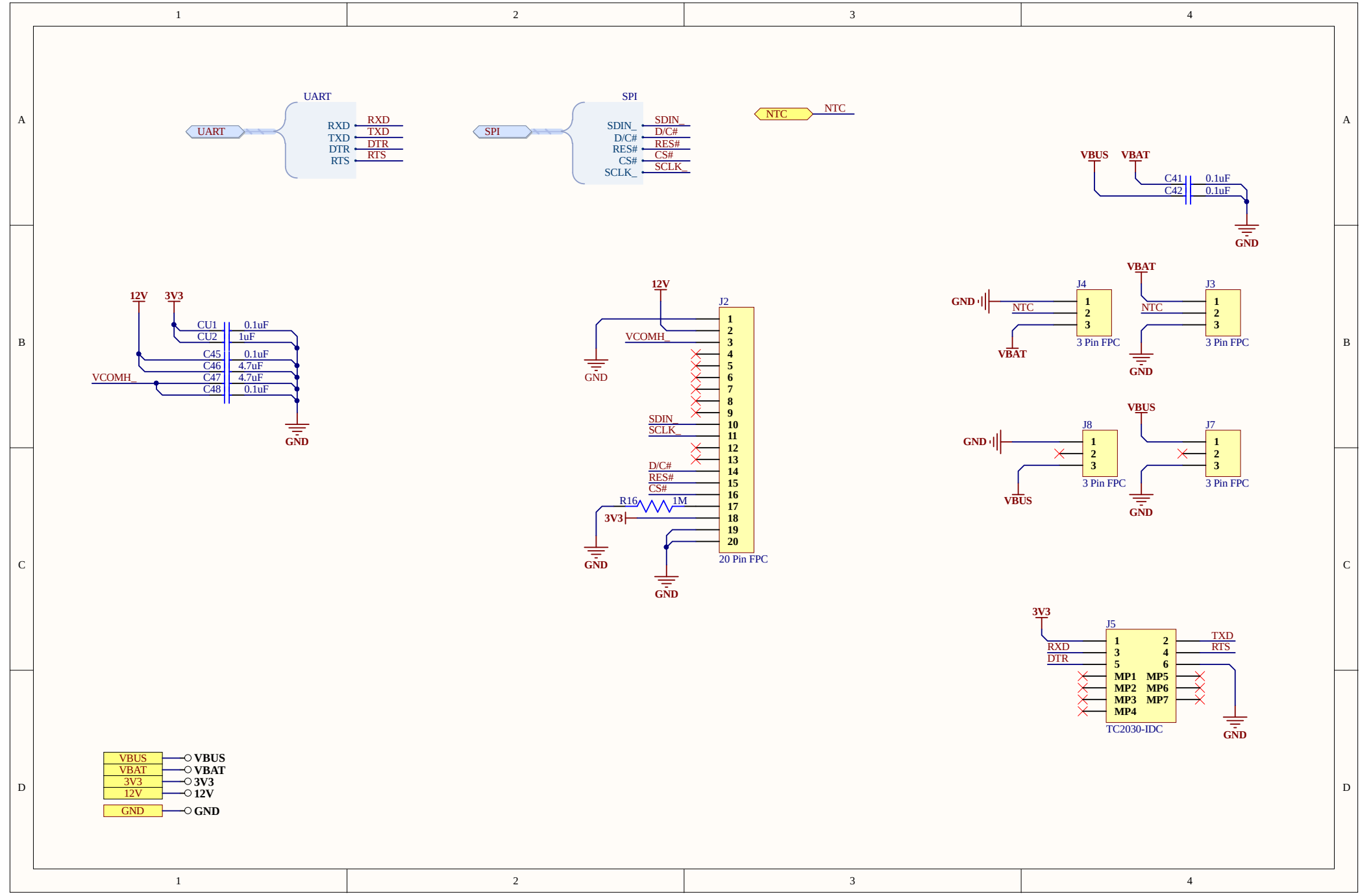

3V3
5V
C32
4.7uF
C33
0.1uF
C34
4.7uF
C35
0.1uF
GND


5V
3V3
U9
8
IN
OUT
1
5
EN
3
NC
2
NC
6
NC
GND
4
7
NC
GND
9
AP7361C-33FGE-7
GND


3V3
VBAT
GND
3V3
VBAT
GND


EN_SIG
PMIC_EN
EN_AD
UART
RXD
TXD
DTR
RTS
TCH_OUT
I2C
INT1
INT2
SDA
SCL
SPI
SDIN_
D/C#
RES#
CS#
SCLK_
J1
FEED
SB1
GND
U.FL-R-SMT-1(80)
3V3
C1 0.1uF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
Two Antenna approaches are used.
One will make it to the final version
AE
L1 2.7nH FEED#
C7 2pF
C8 2.7pF
R1 10K ENABLE
C9 0.1uF
C10 10uF
C11 1uF
C12 1uF
C13 0.1uF
SW1
C14 0.1uF
Switch is present only for testing purpose.
SB2
FEED#
U2A
VDDA
VDDA3P3
VDD3P3_RTC
VDD_SDIO
VDD3P3_CPU
EN
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
U0RXD
U0TXD
XTAL_N_NC
XTAL_P_NC
SD0
SD1
SD2
SD3
CMD
CLK
CAP2_NC
CAP1_NC
LNA_IN
ESP32-PICO-D4
LED_PIN_ R2 1.2K LED1
Q1B UMH3NTN
Q1A UMH3NTN
U2B
IO0
IO2
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
U3
Vcc
CE#
SCLK
Vss
SI/SIO[0]
SO/SIO[1]
SIO[2]
SIO[3]
LY68L6400SLIT


I2C
INT1
INT2
SDA
SCL
TCH_OUT
3V3
R4
10K
SCL
R5
10K
SDA
C15
0.1uF
C16
0.1uF
GND
U4
VDD
VDDIO
AUX_CL
AUX_DA
INT1
INT2
REGOUT
FSYNC
nCS
SDO / AD0
SCL / SCLK
SDA / SDI
RESV
NC
GND
R3
10K
C17
0.1uF
ICM-20649
C18
0.1uF
IC running on Low Power (LP) mode
P2
Test Point
R6
10K
C19
10uF
U5
VDD
SYNC/MODE
SNSK
SNS
OUT
N/C
VSS
AT42QT1011-MAHR


UART
UART
RXD
TXD
DTR
RTS
SPI
SPI
SDIN_
D/C#
RES#
CS#
SCLK_
NTC
NTC
VBUS
VBAT
C41
0.1uF
C42
0.1uF
GND
12V
3V3
CU1
0.1uF
CU2
1uF
C45
0.1uF
C46
4.7uF
C47
4.7uF
C48
0.1uF
VCOMH_
GND
12V
J2
VCOMH_
GND
SDIN_
SCLK_
D/C#
RES#
CS#
R16
1M
3V3
GND
20 Pin FPC
GND
NTC
J4
3 Pin FPC
VBAT
VBAT
NTC
J3
3 Pin FPC
GND
GND
J8
3 Pin FPC
VBUS
VBUS
J7
3 Pin FPC
GND
3V3
J5
RXD
DTR
TXD
RTS
MP1
MP2
MP3
MP4
MP5
MP6
MP7
TC2030-IDC
GND
VBUS
VBAT
3V3
12V
GND
VBUS
VBAT
3V3
12V
GND